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 NVIDIA: Nemotron 3 Ultra (free)[nvidia/nemotron-3-ultra-550b-a55b:free](#)[Model weights](#) [Compare](#)[Playground](#)[Try this model](#)

NVIDIA Nemotron 3 Ultra is an open frontier-reasoning and orchestration model from NVIDIA, with 55B active parameters out of 550B total (MoE). Built on a hybrid Transformer-Mamba mixture-of-experts architecture, it supports text input and ...

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MODALITIES

T → T

PRICE

Free

CONTEXT

1M

RELEASED

Jun 4, 2026

Providers

Different companies host the same model. OpenRouter routes your request to one of them based on the routing mode you pick — Balanced (price + speed), Nitro (fastest), or Exacto (highest tool-calling accuracy).

Free  		Latency / throughput				P50 
Provider 		 Input /M	 Output /M	 Latency	 Throughput	 Uptime
 NVIDIA 		Free	Free	39.55s	4 tps	98.48%

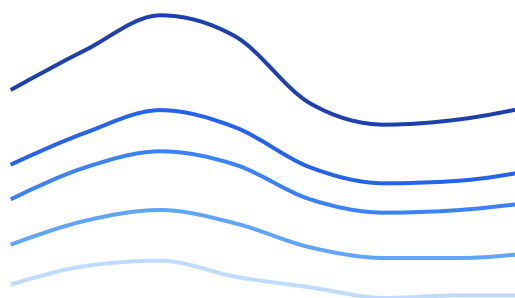
Performance

Throughput is how fast the model writes (tokens per second — higher is better). Latency is total round-trip time (lower is better). TTFT is time-to-first-token — how long before you see anything appear (lower is better).

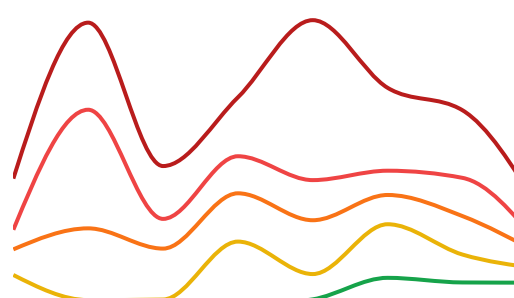
Throughput 4 tok/s P50, best across providers	Latency ⓘ 39.55 s P50, best provider
All locations 	1 week 



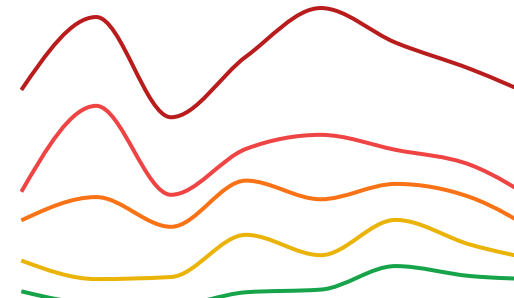
Personal ▾



P99	Avg 90 tok/s
P95	Avg 62 tok/s
P90	Avg 49 tok/s
P75	Avg 30 tok/s
P50	Avg 14 tok/s



P50	Avg 12.21 s
P75	Avg 29.34 s
P90	Avg 52.82 s
P95	Avg 75.16 s
P99	Avg 117.11 s



P50	Avg 39.04 s
P75	Avg 81.55 s
P90	Avg 142.49 s
P95	Avg 196.37 s
P99	Avg 320.80 s

AutoExacto Benchmarks

GPQA DiamondTAU-Bench

Together	81.2%	--
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Tool Call Error Rate



Structured Output Error Rate

BaseTen 85.7% 72.0%

Venice -- 76.0%

Cache Hit Rate



NVIDIA Avg 79.16 %

Uptime is the percentage of the past 3 days that at least one provider was responding to requests.

Availability is the percentage of time that inference was successfully served. OpenRouter continuously monitors and uses the next-best provider when one returns an error.

Uptime (3d)

100.00%

Availability (3d)

76.21%

Availability over the last 3 days Aug 31, 4 PM - Sep 3, 4 PM

Availability 76.21%



Mon

Tue

Wed

Now

Availability over the last 24 hours

100%

90%

70%

50%

30%

Sep 2, 4:40 PM

8:40 PM

12:40 AM

4:40 AM

8:40 AM

12:40 PM

4

When an error occurs in an upstream provider, we can recover by routing to another healthy provider, if your request filters allow it. You can access per-provider uptime data programmatically through the [Endpoints API](#). [Learn more](#) about our load balancing and customization options.

Benchmarks

Scores on standardized evaluations. Higher percentages are better — and rank percentile shows where this model lands among all models on OpenRouter.

 Artificial Analysis Design Arena

38.3

Artificial Analysis
Intelligence Index

Better than 51% of models
compared

49.3

Artificial Analysis
Coding Index

Better than 54% of models
compared

27.5

Artificial Analysis
Agentic Index

Better than 51% of models
compared

Reasoning

GPOA Diamond

Coding

SciCode

HLE

Humanity's Last Exam

28.4%

IFBench

Instruction-following benchmark

81.4%

τ^2 -Bench Telecom

Conversational AI agents in dual-control scenarios

83.3%

AA-LCR

Long context reasoning evaluation

71.0%

GDPval-AA

Economically valuable tasks

33.1%

CritPt

Research-level physics reasoning

3.1%

Terminal-Bench Hard

Agentic coding & terminal use

36.4%

Knowledge

Proportion of correctly answered questions

22.6%

AA-Omniscience Non-Hallucination Rate

Rate of avoiding hallucination among non-correct responses

70.3%

Metrics sourced from [Artificial Analysis](#)

Apps

Public apps that send the most traffic to this model. Good signal for what real production workloads look like — and a hint at which use cases this model is best suited for.

1.		Hermes Agent Hermes Agent is an open-source, self-improving AI agent by Nous Res...	1.64T tokens
2.		Kilo Code Kilo Code is an open-source AI coding agent that works across VS Co...	1.07T tokens
3.		Claude Code Claude Code is Anthropic's agentic coding tool that reads your entire ...	162B tokens
4.		OpenClaw	1.05B tokens

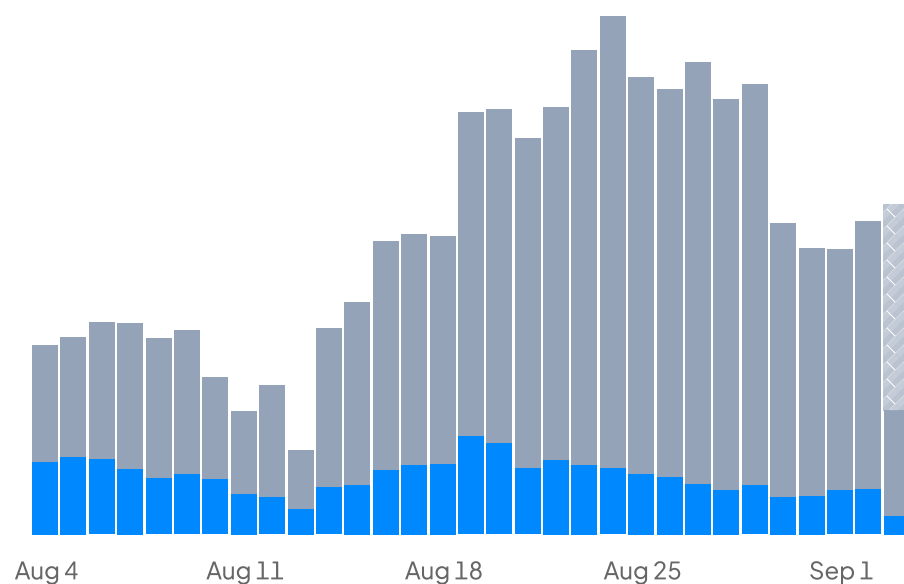
5.



Janitor AI

Janitor AI is a chatbot platform where users create and chat with custo...

103B tokens



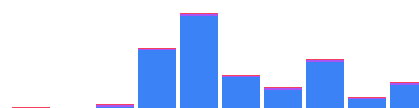
Activity

Token volume and request traffic to this model over time.

Tokens

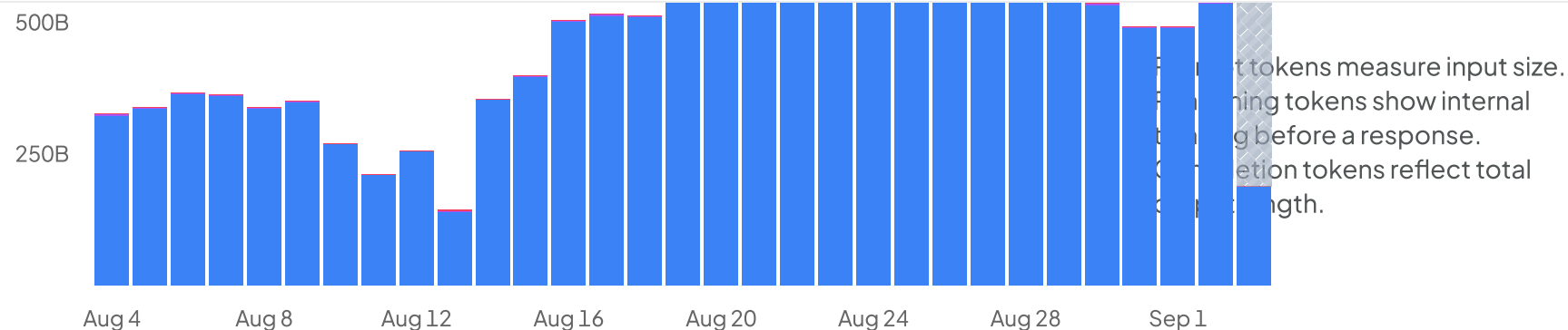
1T

750B



Prompt

188B



Quick Start

Drop-in code to call this model. OpenRouter's API is OpenAI-compatible — most SDKs work by just swapping the base URL. The only thing that changes between models is the model slug below.

1 Get your API key

Create an API key from your OpenRouter dashboard and set it as an environment variable:

Create API Key

```
export OPENROUTER_API_KEY=sk-or-v1- ...
```

[Copy](#)

2 Make your first request

Use `nvidia/nemotron-3-ultra-550b-a55b:free` with the OpenRouter API:

OpenRouter supports reasoning-enabled models that can show their step-by-step thinking process. Use the `reasoning` parameter in your request to enable reasoning, and access the `reasoning_details` array in the response to see the model's

In the examples below, the OpenRouter-specific headers are optional. Setting them allows your app to appear on the OpenRouter leaderboards.

TypeScript SDK Python TypeScript (fetch) cURL Python (OpenAI) TypeScript (OpenAI)

```
import { OpenRouter } from "@openrouter/sdk";

const openrouter = new OpenRouter({
  apiKey: "<OPENROUTER_API_KEY>"
});

// Stream the response to get reasoning tokens in usage
const stream = await openrouter.chat.send({
  chatRequest: {
    model: "nvidia/nemotron-3-ultra-550b-a55b:free",
    messages: [
      {
        role: "user",
```

[Copy](#)

Using third-party SDKs

For information about using third-party SDKs and frameworks with OpenRouter, please see our [frameworks documentation](#).

3 Enable streaming

Add `"stream": true` to your request body to receive responses as server-sent events:

```
curl -N https://openrouter.ai/api/v1/chat/completions \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $OPENROUTER_API_KEY" \
  -d '{
    "model": "nvidia/nemotron-3-ultra-550b-a55b:free".
```

[Copy](#)

```
{ "role": "user", "content": "Hello" }  
]  
'
```

Endpoint

Sends a request for a model response for the given chat conversation. Supports both streaming and non-streaming modes.

POST https://openrouter.ai/api/v1/chat/completions

Authorization	Bearer \$OPENROUTER_API_KEY
Content-Type	application/json
HTTP-Referer	optional – your site URL, for rankings
X-Title	optional – your site name, for rankings
Model	nvidia/nemotron-3-ultra-550b-a55b:free

Creates a streaming or non-streaming response using the OpenAI Responses API format. [Docs](#)

POST https://openrouter.ai/api/v1/responses

Authorization	Bearer \$OPENROUTER_API_KEY
Content-Type	application/json
HTTP-Referer	optional – your site URL, for rankings
X-Title	optional – your site name, for rankings
Model	nvidia/nemotron-3-ultra-550b-a55b:free



POST https://openrouter.ai/api/v1/messages

```
Authorization    Bearer $OPENROUTER_API_KEY
Content-Type     application/json
HTTP-Referer    optional - your site URL, for rankings
X-Title         optional - your site name, for rankings
Model           nvidia/nemotron-3-ultra-550b-a55b:free
```

Parameters

Name	Type	Default	Description
reasoning	map	—	Controls reasoning behavior for models that support thinking tokens, including w
temperature	float	1	This setting influences the variety in the model's responses.
max_tokens	integer	—	This sets the upper limit for the number of tokens the model can generate in respc
seed	integer	—	If specified, the inferencing will sample deterministically, such that repeated requ
top_p	float	0.95	This setting limits the model's choices to a percentage of likely tokens: only the tc
tools	array	—	Tool calling parameter, following OpenAI's tool calling request shape.
tool_choice	string or object	—	Controls which (if any) tool is called by the model.

What is Nemotron 3 Ultra (free)?

Is Nemotron 3 Ultra (free) free?

What is the context length of Nemotron 3 Ultra (free)?

Does Nemotron 3 Ultra (free) support tool calling and structured outputs?

What other text models does Nvidia have?

When was Nemotron 3 Ultra (free) released?

Explore more models

AI Model Rankings 

More models from Nvidia

Nemotron 3.5 ASR Streaming...

Nemotron 3.5 ASR Streaming Multilingual
0.6B is a speech recognition model from ...

Transcription \$0.000003/second

Nemotron 3.5 Lightning

NVIDIA Nemotron 3.5 Lightning is an open
mixture-of-experts model from NVIDIA, ...

Text 262K context \$0.08 / 0.20

Nemotron 3.5 Lightning

NVIDIA Nemotron 3.5 Lightning is an
mixture-of-experts model from NVII

Text 1M context Free



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Labs

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Support

Works With OR

Data

Brand

API Reference

Developer Platform

Status

GitHub

LinkedIn

X

YouTube



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